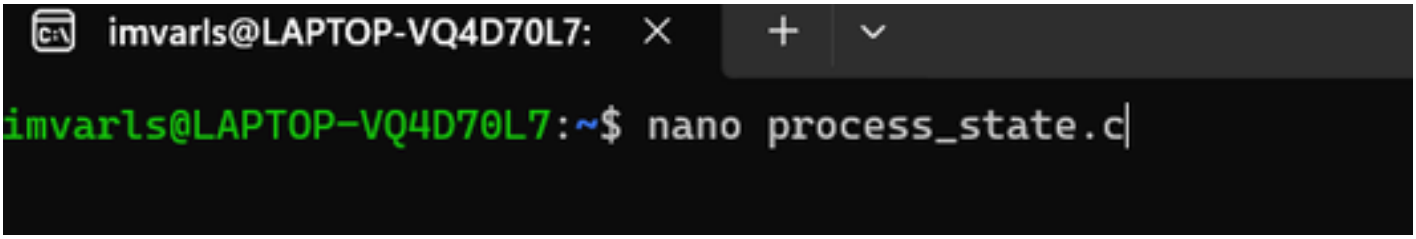
Task 1: Develop a c program using fork() that creates a parent and child process. Display the process ID (PID), Parent process ID (PPID), and process states at different stages of execution

Step 1

Create file



Step 2

imvarls@LAPTOP-VQ4D70L7:

GNU nano 8.7.1 process_state

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("--- Initial Parent Process Stage ---\n");
    printf("Parent PID: %d, Parent's PPID (Shell): %d\n\n", getpid(), getppid());

    // Create a new process
    pid = fork();

    if (pid < 0) {
        perror("Fork failed");
        exit(1);
    }
    else if (pid == 0) {
        // Child Process Block
        printf("[CHILD] Created. PID: %d, PPID: %d\n", getpid(), getppid());

        // Stage 1: CPU-intensive task (Running / Ready state)
        printf("[CHILD] Entering Running/Ready state (Busy loop for 10s)...\n");
        long long i;
        for (i = 0; i < 5000000000LL; i++);

        // Stage 2: Sleep/IO Wait (Waiting / Sleeping state)
        printf("[CHILD] Entering Waiting state (Sleeping for 15s)...\n");
        sleep(15);

        printf("[CHILD] Terminating now.\n");
        exit(0);
    }
    else {
        // Parent Process Block
        printf("[PARENT] Child PID is: %d\n", pid);

        // Stage 1: Parent sleeps while child runs to allow monitoring of the child process
        printf("[PARENT] Sleeping for 35s to allow observation of child state transitions...\n");
        sleep(35);
```

Step 3

imvarls@LAPTOP-VQ4D70L7:

```
imvarls@LAPTOP-VQ4D70L7:~$ nano process_state.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc process_state.c -o process_state
imvarls@LAPTOP-VQ4D70L7:~$ ./file
```

Output

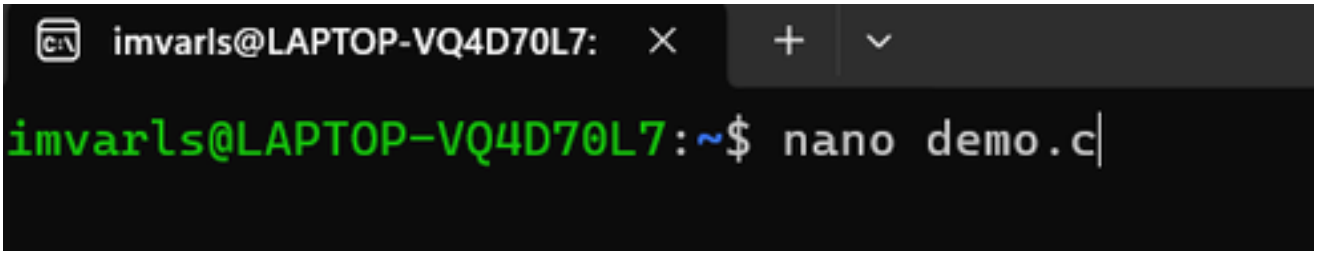
```
imvarls@LAPTOP-VQ4D70L7:~$ nano process_state.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc process_state.c -o process_state
imvarls@LAPTOP-VQ4D70L7:~$ ./file
--- Initial Parent Process Stage ---
Parent PID: 1181, Parent's PPID (Shell): 1120

[PARENT] Child PID is: 1182
[PARENT] Sleeping for 35s to allow observation of child state transitions...
[CHILD] Created. PID: 1182, PPID: 1181
[CHILD] Entering Running/Ready state (Busy loop for 10s)...
[CHILD] Entering Waiting state (Sleeping for 15s)...
[CHILD] Terminating now.

[PARENT] Reaping child process using wait()...
[PARENT] Child reaped. Parent exiting.
imvarls@LAPTOP-VQ4D70L7:~$
```

Task 2 Design an experiment to observe process state transitions (Ready, Running, Waiting, terminates) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

Step 1 Create file



Step 2 Code

```
  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();
    if (pid == 0) {
        // Child process
        printf("Child PID: %d\n", getpid());
        sleep(5); // Waiting state
        for (int i = 0; i < 1e8; i++); // Running state
        exit(0); // Terminated
    } else {
        // Parent process
        printf("Parent PID: %d\n", getpid());
        sleep(10); // Keep parent alive to observe child
    }
    return 0;
}
```

```
imvarls@LAPTOP-VQ4D70L7:
imvarls@LAPTOP-VQ4D70L7:~$ nano demo.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc demo.c -o demo
imvarls@LAPTOP-VQ4D70L7:~$ ./demo
```

Outputs

```
imvarls@LAPTOP-VQ4D70L7:~$ nano demo.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc demo.c -o demo
imvarls@LAPTOP-VQ4D70L7:~$ ./demo
Parent PID: 672
Child PID: 673
imvarls@LAPTOP-VQ4D70L7:~$
```

```
imvarls@LAPTOP-VQ4D70L7:~$ nano demo.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc demo.c -o demo
imvarls@LAPTOP-VQ4D70L7:~$ ./demo
Parent PID: 672
Child PID: 673
imvarls@LAPTOP-VQ4D70L7:~$ ps -o pid,ppid,state,cmd
    PID    PPID S CMD
    581     576 S -bash
    685     581 R ps -o pid,ppid,state,cmd
imvarls@LAPTOP-VQ4D70L7:~$
```

```
imvarls@LAPTOP-VQ4D70L7:~$ top
top - 03:59:36 up 7 min,  1 user,  load average: 0.00, 0.01, 0.00
Tasks:  25 total,   1 running,  24 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7782.6 total,   6493.1 free,    540.0 used,    901.7 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7242.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24232  15376  11584 S   0.0   0.2   0:00.75 systemd
      2 root      20   0    3180   2212   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
      7 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
     47 root      19  -1   50428  16684  15424 S   0.0   0.2   0:00.17 systemd-journal
     79 systemd+  20   0   22540  14616  11996 S   0.0   0.2   0:00.08 systemd-resolve
     85 root      20   0   35364  12352   9260 S   0.0   0.2   0:00.32 systemd-udevd
    144 root      20   0    2888   1948   1820 S   0.0   0.0   0:00.06 chronyd-starter
    145 root      20   0    4472   3160   2908 S   0.0   0.0   0:00.01 cron
    146 message+  20   0    8800   5508   4828 S   0.0   0.1   0:00.07 dbus-daemon
    153 root      20   0   44092  29704  14564 S   0.0   0.4   0:00.38 networkd-dispat
    157 root      20   0   18432   9404   8160 S   0.0   0.1   0:00.11 systemd-logind
    160 root      20   0 1866288  15164  12552 S   0.0   0.2   0:00.13 wsl-pro-service
    183 syslog    20   0  220548   5908   4632 S   0.0   0.1   0:00.12 rsyslogd
    214 _chrony   20   0   24448  11980   7216 S   0.0   0.2   0:00.10 chronyd
    219 root      20   0    5492   2724   2540 S   0.0   0.0   0:00.02 agetty
    223 root      20   0  123456  32592  17980 S   0.0   0.4   0:00.28 unattended-upgr
    232 _chrony   20   0   12092   2444   1796 S   0.0   0.0   0:00.00 chronyd
    330 root      20   0    8644   5556   4716 S   0.0   0.1   0:00.01 login
    372 imvarls   20   0   22332  13516  10900 S   0.0   0.2   0:00.09 systemd
    376 imvarls   20   0   22912   4092   2088 S   0.0   0.1   0:00.00 (sd-pam)
    402 imvarls   20   0    6136   5480   3928 S   0.0   0.1   0:00.01 bash
    575 root      20   0    3184   1124    980 S   0.0   0.0   0:00.00 SessionLeader
    576 root      20   0    3200   1260   1108 S   0.0   0.0   0:00.01 Relay(581)
    581 imvarls   20   0    6268   5536   3864 S   0.0   0.1   0:00.04 bash
    699 imvarls   20   0    8296   6056   3888 R   0.0   0.1   0:00.01 top
```